

EMF & FARADAY'S LAW

Transformer EMF *stationary loop, B changes*

V_{emf} = - \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}

Motional EMF *loop moves, B static*

V_{emf} = \oint (\mathbf{v} \times \mathbf{B}) \cdot d\mathbf{l}

General (flux form) *always works*

V_{emf} = - \frac{d\Phi}{dt} \quad \Phi = \int_S \mathbf{B} \cdot d\mathbf{s}

N-turn coil *transformer, N turns*

V_{emf} = -N \frac{d\Phi}{dt} = - \frac{d\lambda}{dt} \quad \lambda = N\Phi

Uniform B shortcut *B same everywhere on loop*

V_{emf} = - \frac{\partial B}{\partial t} \cdot A

Long wire + rect loop *wire near loop, r: a \to b, height c*

V_{emf} = - \int_0^c \int_a^b \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} = - \int_0^c \int_a^b \frac{\partial}{\partial t} \left(\frac{\mu_0 I}{2\pi r} \right) dr dz

SURFACE ELEMENT ds

d\mathbf{s} = \hat{n} da \quad (RHR gives \hat{n}: curl fingers in loop direction, thumb = \hat{n})

Loop plane	ds
x-y	z dx dy or z r dr d\phi
y-z	x dy dz
x-z	y dx dz
coplanar / toroid	\phi dr dz

Uniform B: \iint da = A (just the loop area)

DOT PRODUCTS \hat{a} \cdot \hat{b}

Same direction = 1; perpendicular = 0

	\hat{x}	\hat{y}	\hat{z}	\hat{r}	\hat{\phi}
\hat{x}	1	0	0	0	0
\hat{y}	0	1	0	0	0
\hat{z}	0	0	1	0	0
\hat{\phi}	0	0	0	0	1

CROSS PRODUCT \mathbf{A} \times \mathbf{B} (determinant)

\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix}

= (A_y B_z - A_z B_y) \hat{x} - (A_x B_z - A_z B_x) \hat{y} + (A_x B_y - A_y B_x) \hat{z}

Also used for curl:

\nabla \times \mathbf{E} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ E_x & E_y & E_z \end{vmatrix}

Plane wave (\partial_x = \partial_y = 0): only \partial_z row survives.

B FIELDS

Infinite wire *single wire carrying I, N = 1*

\mathbf{B} = \hat{\phi} \frac{\mu_0 I}{2\pi r}

Inside toroid *coil of N turns carrying I*

\mathbf{B} = \hat{\phi} \frac{\mu N I}{2\pi r} \quad \mu = \mu_r \mu_0

N in B only if the source (current-carrying thing) is a multi-turn coil. Single wire \Rightarrow N = 1, drop it. N in V_{emf} = -Nd\Phi/dt is the receiver coil turns.

\partial B/\partial t TABLE

B(t)	\partial B/\partial t
B_0 \sin(\omega t)	B_0 \omega \cos(\omega t)
B_0 \cos(\omega t)	-B_0 \omega \sin(\omega t)
B_0 e^{-\alpha t}	-\alpha B_0 e^{-\alpha t}
B_0 (const)	0

SIGN \to CURRENT DIRECTION

V_{emf}	Inside source	I dir.
< 0	- \to + terminal	-\hat{\phi} (CW)
> 0	+ \to - terminal	+\hat{\phi} (CCW)
= 0	—	none

Lenz: induced I always opposes the change in \Phi.

ln INTEGRAL

B \propto 1/r, loop spans r = a to r = b

\int_a^b \frac{dr}{r} = \ln \frac{b}{a} = \ln b - \ln a

\ln \frac{b}{a} = \ln b - \ln a

\ln 2 = 0.693 \quad \ln 3 = 1.099 \quad \ln 1.2 = 0.182

ROTATING LOOP — GENERATOR

\Phi = AB_0 \cos(\omega t + \phi_0)

V_{emf} = AB_0 \omega \sin(\omega t + \phi_0)

I_{amp} = \frac{AB_0 \omega}{R} \quad \omega = 2\pi f = 2\pi \frac{\text{rpm}}{60}

SI PREFIXES

Prefix	Sym	Value	Prefix	Sym	Value
pico	p	10^{-12}	kilo	k	10^3
nano	n	10^{-9}	mega	M	10^6
micro	\mu	10^{-6}	giga	G	10^9
milli	m	10^{-3}	tera	T	10^{12}

j IDENTITIES

Expression	Result	Expression	Result
j^2	-1	j \times (-j)	1
1/j	-j	-1/j	j
j^3	-j	j^4	1
e^{-j\pi/2}	-j	e^{j\pi/2}	j
e^{j\pi}	-1	e^{j2\pi}	1

DISPLACEMENT & CONDUCTION CURRENT

Conduction *resistive, \sigma \neq 0*

E = \frac{V(t)}{d} \quad J_c = \sigma E \quad I_c = J_c \cdot A = \frac{\sigma A}{d} V(t)

I_c = \frac{\sigma A}{1/R} V(t) \iff I = \frac{V}{R} \Rightarrow R = \frac{d}{\sigma A}

Current proportional to V directly \to behaves like a resistor. Displacement *capacitive, changing E*

J_d = \epsilon \frac{\partial E}{\partial t} \quad I_d = J_d \cdot A = \frac{\epsilon_r \epsilon_0 A}{d} \frac{dV}{dt}

I_d = \frac{\epsilon_r \epsilon_0 A}{C} \frac{dV}{dt} \iff I = C \frac{dV}{dt} \Rightarrow C = \frac{\epsilon_r \epsilon_0 A}{d}

Current proportional to dV/dt \to behaves like a capacitor. Lossy cap circuit *both \sigma and \epsilon_r present*

R \parallel C across V(t): I = I_c + I_d = \frac{V}{R} + C \frac{dV}{dt}

CHARGE RELAXATION

Decay *charge injected at t = 0*

\rho_v(t) = \rho_{v0} e^{-t/\tau_r} \quad \tau_r = \frac{\epsilon_r \epsilon_0}{\sigma}

Find \sigma from decay data

\sigma = \frac{-\ln(\rho_v/\rho_{v0}) \cdot \epsilon_r \epsilon_0}{t_1}

Larger \tau_r = slower dissipation = better insulator.

PHASOR METHOD (E \to H)

- Key relation $\mathbf{E}(z, t) = \text{Re}\{\tilde{\mathbf{E}}(z) e^{j\omega t}\}$
- Euler split $e^{j(\omega t - kz)} = e^{j\omega t} \cdot e^{-jkz}$ strip $e^{j\omega t}$, keep e^{-jkz}
- Time \to phasor conversions $\cos(\omega t - kz) = \text{Re}\{e^{j(\omega t - kz)}\} = \text{Re}\{e^{j\omega t} \cdot e^{-jkz}\} \to e^{-jkz}$
 $\sin(\omega t - kz) = \cos(\omega t - kz - \frac{\pi}{2}) = \text{Re}\{e^{j(\omega t - kz - \pi/2)}\} = \text{Re}\{e^{-j\pi/2} \cdot e^{j\omega t} \cdot e^{-jkz}\} \to -j e^{-jkz}$
- Faraday in phasor domain (plane wave: only \partial/\partial z survives)

\tilde{\mathbf{H}} = -\frac{1}{j\omega\mu} \nabla \times \tilde{\mathbf{E}}

- Phasor \to time domain $\mathbf{H}(z, t) = \text{Re}\{\tilde{\mathbf{H}} e^{j\omega t}\}$
 $\text{Re}\{e^{j(\omega t - kz)}\} = \cos(\omega t - kz)$
 $\text{Re}\{-j e^{j(\omega t - kz)}\} = \sin(\omega t - kz)$

UNIT CIRCLE

Angle	Deg	cos	sin	Q
0	0	1	0	—
\pi/6	30	\sqrt{3}/2	1/2	1
\pi/4	45	\sqrt{2}/2	\sqrt{2}/2	1
\pi/3	60	1/2	\sqrt{3}/2	1
\pi/2	90	0	1	—
2\pi/3	120	-1/2	\sqrt{3}/2	2
3\pi/4	135	-\sqrt{2}/2	\sqrt{2}/2	2
5\pi/6	150	-\sqrt{3}/2	1/2	2
\pi	180	-1	0	—
7\pi/6	210	-\sqrt{3}/2	-1/2	3
5\pi/4	225	-\sqrt{2}/2	-\sqrt{2}/2	3
4\pi/3	240	-1/2	-\sqrt{3}/2	3
3\pi/2	270	0	-1	—
5\pi/3	300	1/2	-\sqrt{3}/2	4
7\pi/4	315	\sqrt{2}/2	-\sqrt{2}/2	4
11\pi/6	330	\sqrt{3}/2	-1/2	4
2\pi	360	1	0	—

FIELDS

E — electric field (V/m)
B — mag. flux density (T)
H — mag. field intensity (A/m)
\rho_v — charge density (C/m^3)
\rho_{v0} — initial charge density
\Phi = \int_S \mathbf{B} \cdot d\mathbf{s} — flux (Wb)
\lambda = N\Phi — flux linkage (Wb)

CURRENTS

J_c = \sigma E — cond. density (A/m^2)
J_d = \epsilon \partial E/\partial t — disp. density
I_c = J_c \cdot A — cond. current (A)
I_d = J_d \cdot A — disp. current (A)
R = d/(\sigma A) — resistance (\Omega)
C = \epsilon_r \epsilon_0 A/d — capacitance (F)
\tau_r = \epsilon_r \epsilon_0/\sigma — relax. time (s)

MATERIALS

\mu_0 = 4\pi \times 10^{-7} \text{ H/m}

\mu_r — relative permeability
\mu = \mu_r \mu_0 — permeability
\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}

\epsilon_r — relative permittivity
\epsilon = \epsilon_r \epsilon_0 — permittivity
\sigma — conductivity (S/m)

CIRCUIT & WAVE

V — voltage (V)
R — resistance (\Omega)
C — capacitance (F)
\omega = 2\pi f — angular freq (rad/s)
k = \omega\sqrt{\mu\epsilon} — wave number (m^{-1})
\lambda = 2\pi/k — wavelength (m)
c = 3 \times 10^8 \text{ m/s} — speed of light

PHASOR & GEOMETRY

\eta = \sqrt{\mu/\epsilon} — intrinsic imp. (\Omega)
\eta_0 = 120\pi \approx 377 \Omega (free space)
k/(\omega\mu) = 1/\eta
j — imaginary unit (j^2 = -1)
\tilde{\mathbf{E}} — phasor (drop e^{j\omega t})
A — area (m^2); d — sep. (m)
N — turns; r — radial dist. (m)